

Flow measurement in streams using video imagery

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Received 18 March 2002; revised 27 August 2002; accepted 27 August 2002; published 27 December 2002.

[1] A video imagery technique for making flow measurements in streams and waterways is presented and used to estimate discharge for Clear Creek near Oxford, Iowa. A video camera was used to visualize the flow seeded with tracers. Measurements of free surface flow velocities were then made using particle image velocimetry (PIV) techniques. The velocity measurements were used as input to a hydraulic model, which uses kinematic principles (conservation of mass) to derive three-dimensional flow fields for discharge estimation. A survey of five channel cross sections over a 7.15 m length of stream were made to define the channel bathymetry for the hydraulic model. The average channel width was 5.7 m, and the average depth was 0.2 m. The discharge estimate using the video technique ($0.183 \text{ m}^3/\text{s}$) compares well with the current meter discharge measurement ($0.192 \text{ m}^3/\text{s}$ with an estimated standard error of 6.4%). The root-mean square difference of the depth-averaged velocity at the locations of the current meter measurements was 0.032 m/s. Potential applications of the video imagery technique include measurements of flood flows at ungaged sites, river monitoring from a remote site, and estimation of two- and three-dimensional flow components.

INDEX TERMS: 1821 Hydrology: Floods; 1824 Hydrology: Geomorphology (1625); 1860 Hydrology: Runoff and streamflow; 1894 Hydrology: Instruments and techniques; **KEYWORDS:** particle image velocimetry, discharge measurement, remote sensing, stream gaging

Citation: Bradley, A. A., A. Kruger, E. A. Meselhe, and M. V. I. Muste, Flow measurement in streams using video imagery, *Water Resour. Res.*, 38(12), 1315, doi:10.1029/2002WR001317, 2002.

1. Introduction

[2] Flow measurements are an integral part of stream gaging. At U.S. Geological Survey (USGS) stream gages, discharge measurements are usually made several times a year to maintain flow rating curves, which relate discharge to river stage [Wahl *et al.*, 1995]. For over a hundred years, current meters have been used to measure river discharge. The approach is well-documented [e.g., Buchanan and Somers, 1969] and accurate [Pelletier, 1988; Sauer and Meyer, 1992]. However, it does require contact with the flow, which can be hazardous to people or equipment during flood measurements.

[3] Over the past decade, new techniques have played a larger role in flow measurement. Advanced electronic sensing equipment, based on acoustic Doppler velocimetry (ADV), have been used to measure one-, two-, and three-dimensional flow fields in natural waterways. Although discharge measurements can be made quickly with ADV, contact with the flow is still required, and the equipment is relatively expensive to purchase and operate. Still, ADV has

proven to be extremely valuable for measurement of complicated flow patterns [Oberg and Schmidt, 1994].

[4] In recent years, remote sensing technologies have also been investigated for river monitoring. Doppler radar measurements of surface currents, made from the river's edge, have been tested for discharge estimation [Costa *et al.*, 2000]. Remote measurements of river channel bathymetry have been demonstrated with a ground penetrating radar [Spicer *et al.*, 1997; Costa *et al.*, 2000]. Satellite-based radar altimetry has also been used to estimate water levels for large rivers [Koblinsky *et al.*, 1993; Vörösmarty *et al.*, 1996; Birkett, 1998] and in flooded river valleys [Brakenridge *et al.*, 1998]. The combination of satellite remote sensing measurements with hydraulic modeling can provide unique information for hydrologic and geomorphic studies [Mertes, 1994].

[5] In this paper, we describe a new remote sensing approach for flow measurement in rivers and streams. The approach involves two steps: (1) surface velocity measurement using video imagery techniques, and (2) discharge estimation using a kinematic flow analysis. Surface velocity measurements are made using particle image velocimetry, a well established technique in fluids research for laboratory flow measurement. Discharge and other flow variables are estimated using a kinematic hydraulic model that utilizes the

surface velocity measurements in a iterative estimation procedure. A field application at Clear Creek near Oxford, Iowa, illustrates the proposed video imagery techniques.

2. Image Velocimetry

[6] In recent years, image velocimetry has become a popular method for two- and three-dimensional flow measurement for laboratory-scale fluids experiments [Adrian, 1991; Willert and Gharib, 1991; Buchhave, 1992]. An appealing aspect of image velocimetry is its inherent simplicity. First the flow is seeded with tracer particles that follow the fluid movement accurately. Then, images of the region(s) of interest are recorded. If the tracer particle density is low, one can determine the displacement of individual particles between successive frames. Velocity is then calculated by dividing particle displacement by the time interval between successive frames. This technique is known as particle tracking velocimetry (PTV). If the particle density is high, then the movement of groups of particles is determined using a statistical approach. This technique is known as particle image velocimetry (PIV).

[7] The requirements for flow measurement using PIV are a series of images containing flow tracers, a method of digitizing the images, and analysis software for velocity estimation. The images may be comprised of a series of photographs, or more conveniently, images from a video camera. In this latter case, the images must be digitized with a personal computer equipped with a frame-grabber. The tracers must be large enough so that they map to at least one pixel, but they must be small and light enough to accurately trace the free surface movements. In order to obtain accurate velocity estimates a large number of tracers are required. In addition, they must form sufficient contrast with the water surface. In a field setting, controlling these factors is difficult. During high flows, velocity measurement can often be made from naturally occurring tracers, including foam, boil and ripple structures, and debris. However, during low flow measurement, the flow may need to be “seeded” to obtain sufficient tracers.

[8] While PIV is conceptually simple, there are many small details that must be resolved in order to get reliable velocity measurements. Careful implementation of an algorithm is important. The PIV analysis software used in this study was developed over many years by I. Fujita (Gifu University, Japan) and modified for large-scale applications at IIHR-Hydroscience and Engineering. The code is solid and very well-tested in several prior field and laboratory applications [Ettema et al., 1997; Fujita et al., 1998; Aya et al., 1995]. In the next sections, the theory behind velocity estimation using video imagery is discussed.

2.1. PIV Technique

[9] Two major approaches are used in PIV for motion estimation from a sequence of images: (1) differential techniques, and (2) correspondence techniques. In differential methods one models the image sequence with a three-dimensional function $f(x,y,t)$. In correspondence methods one searches for a direct correspondence (correlation) between a group of pixels between frames. There are many variants and correspondence methods are sometimes called matched-filter or template methods. All of these approaches have been used in earth science applications [Leese et al.,

1971; Ninnis et al., 1986; Emery et al., 1986; Beck and Plaskowski, 1987; Collins and Emery, 1988; Utami and Ueno, 1991; Stevens and Coates, 1994; Lloyd et al., 1995].

[10] Although it is not immediately clear which class of methods produce the most accurate results, Abidi and Gonzalez [1987] and Cowen and Monismith [1997] have demonstrated the superiority of correspondence methods in terms of accuracy, while differential methods are computationally efficient. The PIV method utilized here is a variation on the classical PIV correspondence method.

[11] Figure 1 shows two consecutive flow fields overlaid with an imaginary grid. An interrogation spot is selected around the center of a grid point in the first field. Next, one searches for the most likely location of this group of pixels in the second field as follows. An enclosing search area is selected around the corresponding grid point in the second field. The interrogation spot is placed in the upper left hand corner of the search area, and the linear correlation coefficient R between the two sets of pixels is computed as:

$$R = \frac{\sum_l \sum_k (a_{kl} - \bar{a}_{kl})(b_{kl} - \bar{b}_{kl})}{\sqrt{\sum_l \sum_k (a_{kl} - \bar{a}_{kl})^2 \sum_l \sum_k (b_{kl} - \bar{b}_{kl})^2}} \quad (1)$$

where a_{kl} are the grey-scale values for pixels in the interrogation spot, and b_{kl} are the grey-scale values for the corresponding pixels in the search area. The interrogation spot is then moved right one position and R is computed again. This procedure is repeated for all the possible locations of the interrogation spot in the search area. The location where the correlation coefficient takes on its highest value is taken as the most likely location of the pixels from the interrogation spot in the search area. With the displacement and time difference between fields now known, one can compute an estimated velocity at the specific grid point. The whole process is repeated at every grid point and the result is a field of velocity vectors.

[12] The quality of the results from the PIV analysis strongly depends on the choice of the location and size of the interrogation spot and search area, and the size and shape of the imagery grid (see Figure 1). For example, if the interrogation spot is too small, many sets of pixels in the search area may give high correlation, leading to erroneous vectors. The same is true if the search area is too big. On the other hand, if the interrogation spot is too big, the differences between the computed correlation coefficients become small, also resulting in erroneous vectors. Extensive numerical simulations have been done on how to select the analysis parameters (interrogation spot, search area, grid spacing) for experiments in fluid mechanics. For a discussion of this work, and guidance on parameter selection for PIV analysis, see Adrian [1991].

2.2. Image Registration

[13] To measure velocities from an image sequence one has to relate pixel locations to physical locations. In the laboratory this step is straightforward. A grid with known dimensions is placed in the flow facility. Since experiments are almost always designed so that the camera is perpendicular to the imaged area, the problem is inherently two-dimensional. Experiments typically require a small field of

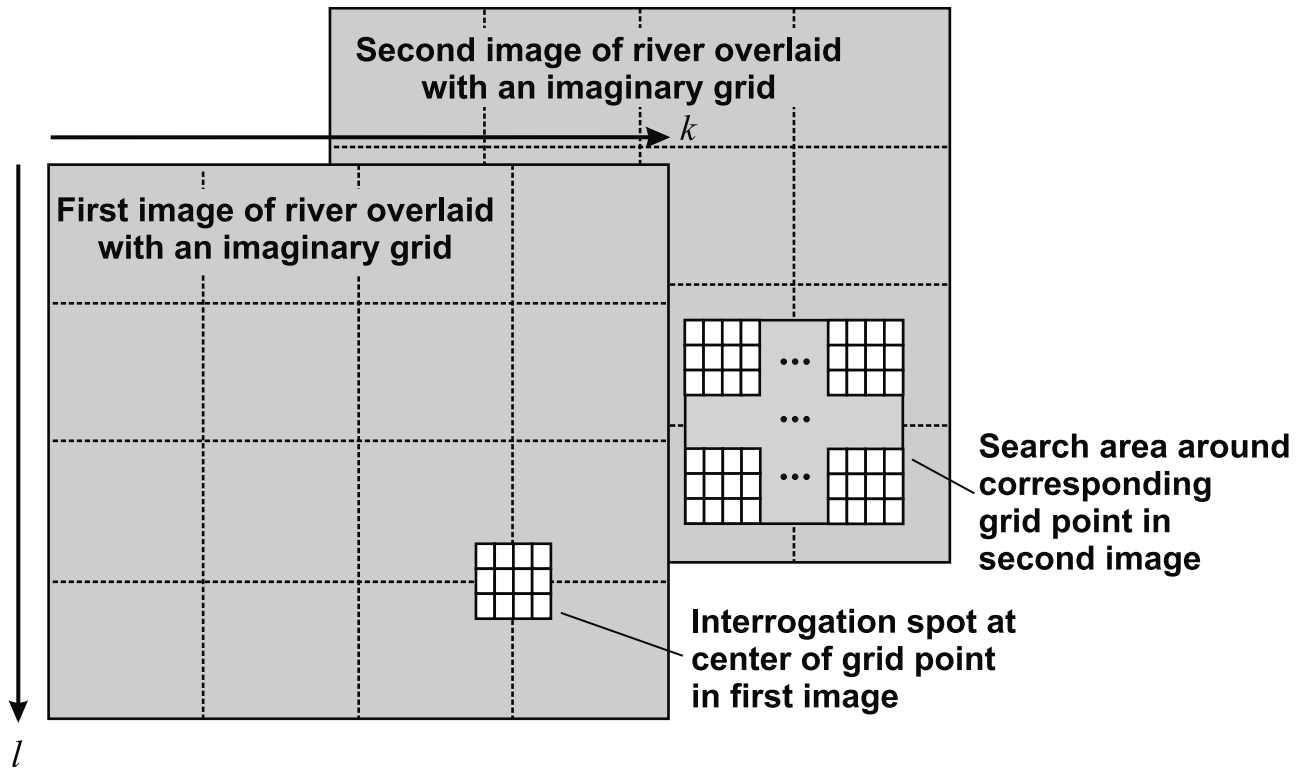


Figure 1. Particle image velocimetry. Using two consecutive images of the flow field, the pattern in an interrogation spot on the first image is compared to the patterns in a search area of the second image. For each possible location in the search area, the linear correlation coefficient between the pixels from the interrogation spot and the corresponding pixels in the search area is computed.

view, normally less than 5–7 degrees, so the camera lens introduces little spatial distortion. Oftentimes, so-called telecentric lenses, or lenses that preserve spatial relationships, are used. Therefore a two-dimensional (bilinear) interpolation can be used to estimate the location of a pixel in physical space.

[14] In a field application, it is usually impossible to image the water surface from directly overhead. Instead, the video camera views the water surface from an oblique angle. If the field of view is large, the camera lens may introduce spatial distortion of the image. A three-dimensional coordinate transformation is required to map the image coordinates into physical coordinates. This step requires selection of a few surveyed control points on the perimeter of the imaged area with known physical coordinates.

[15] A general approach for image registration is the use of a three-dimensional conformal coordinate transformation [Wolf, 1983]. A simpler, more direct approach, based on an eight-parameter projective transformation, has been used by Fujita *et al.* [1998] to estimate the physical location of points from the image in field applications. The transformation has the form:

$$Y_1 = \frac{a_1x_1 + a_2x_2 + a_3}{a_7x_1 + a_8x_2 + 1} \quad (2)$$

$$Y_2 = \frac{a_4x_1 + a_5x_2 + a_6}{a_7x_1 + a_8x_2 + 1} \quad (3)$$

where (x_1, x_2) is the point in image coordinates, (Y_1, Y_2) is the point in physical coordinates, and a_i are the transformation parameters. This fractional linear transformation can empirically model the distortion effects for oblique images [Mikhail and Ackerman, 1976]. However, the approach assumes a horizontal water surface, and requires control points on the horizontal surface for parameter estimation. A minimum of four control points are required to obtain a solution. If additional points are available, the transformation equations can be written as a linear combination of the parameters a_i , and least squares can be used to estimate the parameters. Since the size of a pixel in physical space is usually much larger than the uncertainty in the control point location (i.e., based on standard surveying techniques), four control points are sufficient. However, additional control points are recommended, since any errors in the control point survey could be detected, and the least squares analysis could provide information on the uncertainty of the image transformation.

3. Discharge Estimation Using PIV

[16] Video images of tracers on the surface provide measurements of free surface velocities at points within the imaging area. To effectively utilize these data in discharge estimation, a numerical model is needed to assimilate the measurements under imposed physical constraints. Kinematic flow analysis is a robust and computationally inexpensive numerical tool that can be used to analyze complex flow patterns. With this approach, a numerical

model generates a mass-conserving three-dimensional flow field using a discrete set of surface measurements, specified boundary conditions, and bathymetry data, based solely on kinematic (continuity) principles. Hence, the kinematic model serves as a physically-based interpolation scheme for assimilating surface velocity measurements in flow field estimation.

3.1. Kinematic Flow Analysis

[17] The numerical model used for discharge estimation solves the three-dimensional continuity equation for an incompressible fluid:

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0, \quad (4)$$

where u , v , and w are the scalar components of the velocity in the x , y , and z directions. The model does not solve the momentum equation, and therefore it accounts only for the flow kinematic properties. However, for an incompressible fluid, all the kinematic properties of the flow are directly related to the flow velocity components.

[18] The kinematic analysis is initialized with the surface velocities measured using PIV. The flow is assumed to be zero on the channel banks and bed (i.e., no-slip condition), and a zero flux boundary condition is imposed at the free surface. The upstream and downstream boundaries can be initialized in several ways (e.g., uniform, logarithmic, or power law velocity profiles). The numerical model initially interpolates the measurements and boundary conditions over the remainder of the uniform Cartesian grid encompassing the flow domain under consideration. The continuity equation is then solved iteratively using a successive over-relaxation (SOR) technique until a steady state mass-conserving flow field is obtained [Stronach *et al.*, 1993; Varga, 2000]. SOR is a numerical technique used to accelerate the convergence toward the solution of a system of linear equations.

3.2. Discharge Estimation

[19] Discharge estimation is an iterative procedure. Using the measured surface velocities to initialize the numerical model, the kinematic analysis is carried out to obtain a steady state mass-conserving flow field for an assumed discharge Q . Next, the differences between the free surface velocities determined by kinematic analysis and those from the video imagery measurements are computed. This was carried out in this study using a mean absolute difference objective function:

$$F(Q) = \frac{1}{N} \sum_{i=1}^N |\mathbf{V}_{m_i} - \mathbf{V}_{k_i}(Q)| \quad (5)$$

where $\mathbf{V}_{m_i}$ is the vector velocity based on the video imagery measurement at point i , $\mathbf{V}_{k_i}(Q)$ is the vector velocity based on the kinematic analysis at point i for the assumed discharge Q , and N is the number of available points with velocity measurements. The assumed discharge Q is then adjusted and the process is repeated until the minimum of $F(Q)$ is found. The discharge corresponding to the minimum is taken to be the discharge estimate. Such a

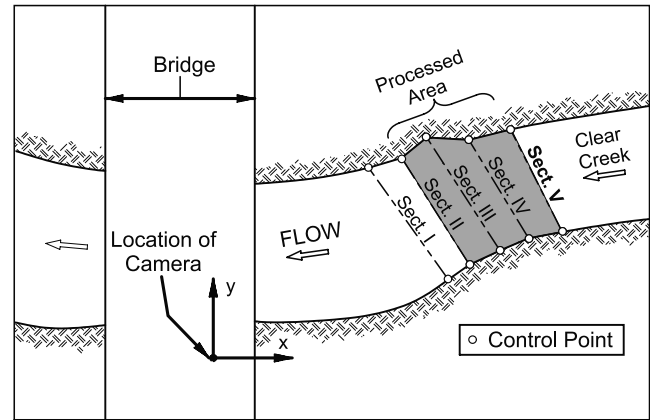


Figure 2. Study site at Clear Creek near Oxford, Iowa.

procedure could be automated using a search algorithm based on the objective function.

4. Case Study

[20] Initial field experiments were carried out to evaluate the video imagery technique for flow measurement. The experiment was carried out on 17 October 1997 at the USGS Gage 05454220 Clear Creek near Oxford, Iowa (drainage area of 58.4 mi²). Figure 2 shows the Clear Creek site. Tag lines were used to establish five channel cross sections for the site survey control. Measurements of water depths were made at each cross section to estimate the channel bathymetry for the kinematic model.

4.1. Video Measurements

[21] Video images were recorded from a bridge (see Figure 2) using a Panasonic Palmcorder PV IQ405 video camera. For the low flow conditions at the time of the measurement, insufficient natural tracers were available to visualize the flow. Therefore flow seeding was required. Fallen leaves (gathered before the experiment) were spread across the channel cross section upstream of the imaging area. The video camera recorded the passage of the leaves through the imaging area for a 3-minute period.

4.2. Image Processing

[22] The first step of PIV image processing is the digitization of the video images. The video camera recorded images at a rate of 30 frames per second. A total of 60 images were digitized with a 1-s interval between images. The digitizing system consisted of a PC with a framegrabber and a computer-controlled VCR. After framegrabbing the images were enhanced by performing histogram equalization. This is a simple but very effective procedure that can dramatically enhance the contrast in an image [Russ, 1995].

[23] The next step is of the calculation of pixel displacement vectors using the PIV technique. Displacement vectors were computed for 30 image pairs with the correlation method using the parameters shown in Table 1. Analyzing image sequences to extract velocity fields is computationally expensive. Running time depends linearly on the number of images and quadratically on the desired accuracy. To analyze the 60 images on a 200 MHz Pentium PC required about 2 hours. Afterward, the algorithm developed

Table 1. Parameters for PIV Analysis of Clear Creek

Parameter	Value
Number of images	60
Time between images	1 s
Grid spacing	32 pixels
Interrogation spot	25 × 25 pixels
Search area size	48 × 48 pixels

by Fujita and Kaizu [1995] was used to detect and correct erroneous vectors produced by the statistical procedure. Average velocity vectors were then computed at each point. Figure 3 shows the averaged pixel displacement vectors overlaid on one of the grabbed images. Vectors outside the region of interest were masked out.

[24] The final image processing step is the transformation of the pixel displacement vectors into an undistorted space with known physical coordinates and dimensions. The 8-parameter transformation shown in equations (1) and (2) was used. Ten points along the edge of the stream were used for site control and the parameters of the transformation were estimated by least squares. The physical coordinates of the control points were determined by a site survey using a Topcon GTS-28 total-station theodolite with distance measuring equipment. Figure 4 shows the estimated surface velocity vectors.

4.3. Kinematic Model Discharge Estimates

[25] The code used in this analysis was developed and tested in previous applications [Hay *et al.*, 1997]. The computational grid used for the kinematic model has a Δx and Δy of 0.1542 m (0.5 ft), while Δz was chosen to be 0.03048 m (0.1 ft). Surface velocities for the computation grid were initialized by interpolating velocity vectors from the PIV analysis. A logarithmic velocity profile, corresponding to fully turbulent flow conditions, was assumed at both the upstream and downstream boundaries. Comparisons of the kinematic model outputs and the PIV surface velocity measurements were made to evaluate the mean absolute difference in the magnitude of the velocity vectors. Figure 5

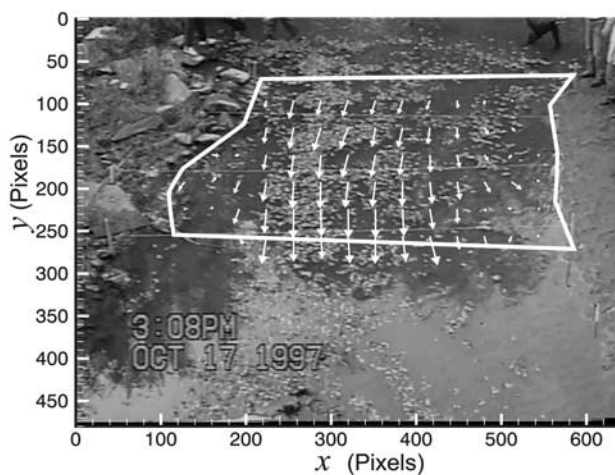


Figure 3. Average pixel displacement vectors (in pixels/s) overlaid on one of the digitized images. The average is based on 60 images taken at a 1 s interval.

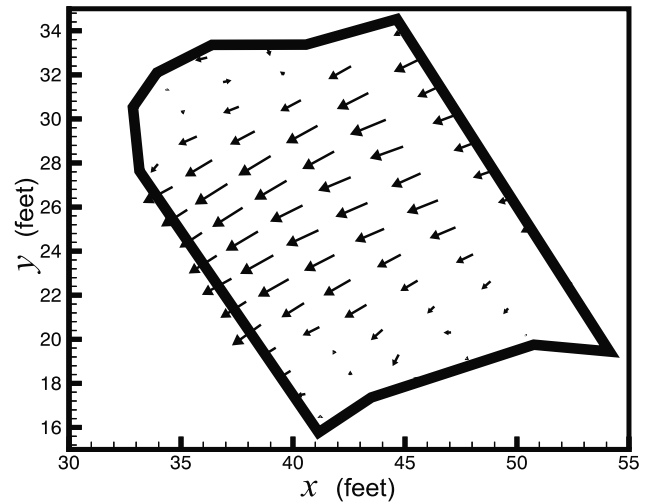


Figure 4. Estimated surface velocity vectors (in ft/s; 1 foot = 0.3048 m) in physical coordinate space.

shows the mean absolute difference $F(Q)$ versus discharge. The discharge estimate is $0.183 \text{ m}^3/\text{s}$, which corresponds to the minimum mean absolute difference of 0.054 m/s.

[26] For the observed flow conditions, the Reynolds number is about 4×10^6 , so the assumption of fully turbulent flow is reasonable for the boundary conditions. However, to examine the sensitivity of discharge estimates to this assumption, a second kinematic analysis was per-

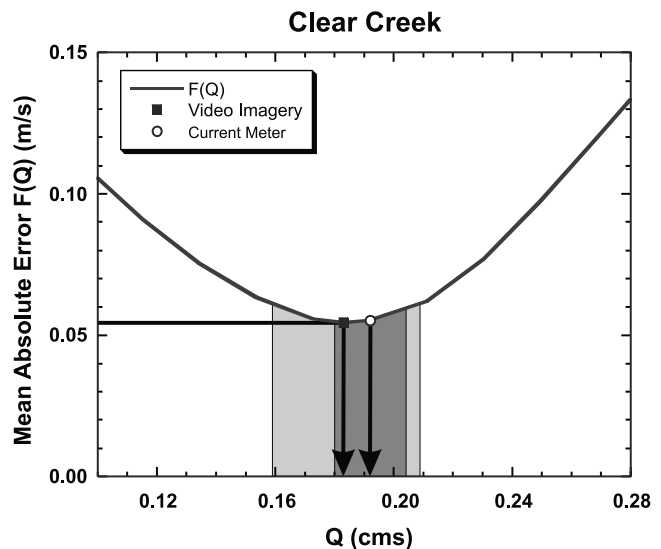


Figure 5. Mean absolute difference $F(Q)$ (in m/s) between the PIV surface velocity estimates and the simulated velocities by the kinematic model as function of discharge. The kinematic model was run assuming a logarithmic velocity profile for upstream and downstream boundary conditions. The final discharge estimate (corresponding to the minimum) is $0.183 \text{ m}^3/\text{s}$. The current meter measurement (also shown) is $0.192 \text{ m}^3/\text{s}$. The light shaded area shows the (approximate) range within one standard error of the final discharge estimate. The dark shaded area defines the range within one standard error of the current meter measurement.

Table 2. Current Meter Measurements of Discharge for Clear Creek Near Oxford, Iowa, at 2 PM CDT 17 October 1997

Section	Distance, m	Effective Width, m	Depth, m	Area, m ²	Velocity, m/s	Discharge, m ³ /s
0	0.000	0.254	0.000	0.000	0.000	0.000
1	0.508	0.508	0.165	0.084	0.116	0.010
2	1.016	0.508	0.305	0.155	0.163	0.025
3	1.524	0.508	0.400	0.203	0.228	0.046
4	2.032	0.508	0.280	0.142	0.205	0.029
5	2.540	0.508	0.315	0.160	0.196	0.031
6	3.048	0.508	0.250	0.127	0.154	0.020
7	3.556	0.508	0.208	0.106	0.180	0.019
8	4.064	0.381	0.150	0.057	0.114	0.007
9	4.318	0.635	0.125	0.079	0.065	0.005
10	5.334	0.508	0.000	0.000	0.000	0.000
Total		5.334		1.113		0.192

formed assuming uniform velocity profiles for the upstream and downstream boundaries. The results for the uniform profile case were nearly identical to those for the logarithmic profile case. The discharge estimate is $0.184 \text{ m}^3/\text{s}$, and the curve of $F(Q)$ versus discharge for the uniform profile case is indistinguishable from the one shown in Figure 5. Therefore, in this experiment, the results are insensitive to the assumption made for the upstream and downstream boundary conditions.

[27] Although methods for evaluating the uncertainty of the video-imagery discharge estimate have not yet been developed, it is possible to obtain a rough estimate. This was done by examining the sensitivity of $F(Q)$ to the uncertainty in the PIV velocity measurements. If one assumes that the errors in the PIV velocity measurements are independent, with a standard error equal to 20% of the measured velocity (a conservative guess), a first-order approximation of the standard deviation of $F(Q)$ can be obtained. Values of $F(Q)$ within one standard deviation of the minimum correspond to discharges within about 13.6% ($\pm 0.025 \text{ m}^3/\text{s}$) of the discharge estimate. As expected, this uncertainty is larger than the standard error usually observed for current meter measurements, and results because a range of discharges yield values of $F(Q)$ close to the minimum (see Figure 5).

4.4. Current Meter Discharge Measurements

[28] For comparison, a discharge estimate was made using conventional current meter measurements along tag line 4 (see Figure 2). To the extent possible, we followed USGS guidelines for discharge measurement [Buchanan and Somers, 1969]. However, for low flow measurement on small streams, the narrow channel width and shallow water depths usually make it impractical to follow recommended guidelines regarding the minimum number of vertical profiles for the cross section, and minimum number of depths for velocity measurements at each profile (K. Oberg, personal communication).

[29] An A. Ott Kempton Type C-2 current meter was used for velocity measurements. Measurements were made for nine vertical profiles. The average spacing between profiles was 0.53 m. Wherever possible, velocity measurements were made at 0.2, 0.6, and 0.8 of the water depth at the profile. Average profile velocities were then estimated using the three-point measurements. However, at some profiles (near the channel banks), water depths were too low to

permit velocity measurements at all depths. Therefore average profile velocity estimates based on one- or two-point measurements were used. Table 2 shows the discharge computations for the cross section. The estimated discharge is $0.192 \text{ m}^3/\text{s}$. The uncertainty in the discharge estimate was evaluated using the procedures developed by Sauer and Meyer [1992]. However, instrument error curves for a standard rated Price Pygmy meter were used because instrument error estimates for the current meter used in this study are not available. The estimated standard error of the discharge measurement is 6.4% ($\pm 0.012 \text{ m}^3/\text{s}$).

4.5. Comparison

[30] Discharge and velocity estimates based on the video imagery approach ($0.183 \text{ m}^3/\text{s}$) are remarkably similar to those based on the current meter measurements ($0.192 \text{ m}^3/\text{s}$). The difference is within the $\pm 0.012 \text{ m}^3/\text{s}$ estimated standard error for the current meter measurement (see Figure 5). Figure 6 shows the depth average velocity estimates for

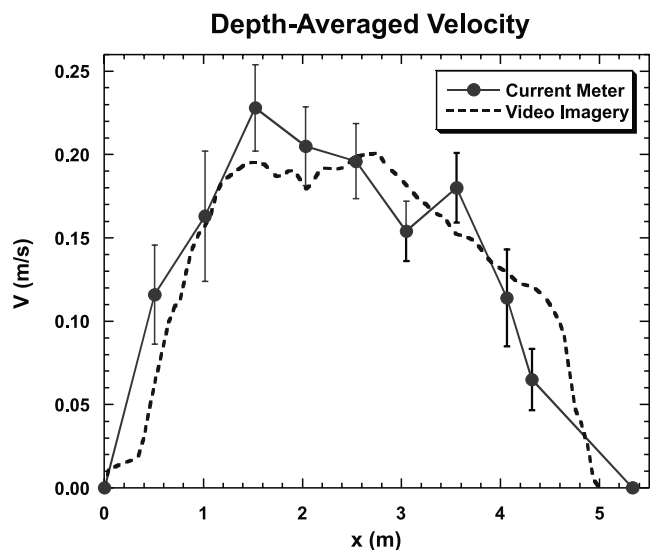


Figure 6. Depth-averaged velocity estimates based on current meter measurements (points) and the video imagery approach (line). Uncertainties in the current meter measurements (two standard deviations) are indicated by the bars. Video imagery results are based on the kinematic model analysis for the logarithmic velocity profile case.

the cross section 4 for the current meter measurements and results of the video imagery approach (logarithmic profile case). Also shown are the estimated standard errors for current meter measurements using procedures adapted from *Sauer and Meyer* [1992]. The differences between velocity estimates are less than 0.06 m/s. The root-mean-square difference for the 9 points within the stream channel are 0.032 m/s. Hence, estimates based on the video imagery techniques are consistent with conventional measurement techniques.

[31] Provisional discharge estimates were also available in real-time for the USGS gage located approximately 15 m downstream of the Clear Creek site. The provisional estimate for the time of the measurement was 0.085 m³/s, or about 44% of the current meter measurement. The final published daily-average discharge for this date, which may account for shifts in the rating curve, is 0.091 m³/s. These large observed differences illustrate the significant errors that can result for low flow estimates based on traditional stage-discharge rating curves.

5. Concluding Remarks

[32] A video imagery technique has been demonstrated for use in flow measurement. The approach involves (1) surface velocity measurement, using field PIV techniques, and (2) discharge estimation, using kinematic flow analysis. Equipment costs for flow measurement are relatively modest. Video measurements require a video camera, a video recorder, and a frame-grabber. A PC computer is needed to run the PIV and kinematic analyses. As with conventional current meter discharge measurements, a site survey is also needed to define channel cross sections. However, after the initial survey, the approach could be applied again with little or no site preparation (for a stable section).

[33] This initial evaluation of the feasibility of making flow measurements in natural waterways using video imagery identified unique circumstances that will impact the quality of field video measurements. Among the most challenging problems is the seeding of the flow in a field setting. In some instances (e.g., flood flows), sufficient natural tracers are available for flow visualization. However, for the low flow conditions at Clear Creek, seeding of the flow with tracers was required. We are currently exploring alternatives for flow seeding in field applications, as well as the use of alternative video image processing techniques, which might reduce or eliminate the need for flow seeding. Another limitation is the assumption of a planar water surface for image transformation. In steep river channels in mountainous terrain, or during high flows at some locations, the water surface can exhibit strong undulations. Making quantitative measurements using video in these situations might be accomplished using a pair of cameras, which could provide stereographic images for photogrammetric analysis.

[34] Obtaining channel cross section information is also a significant concern. Measurement of channel cross sections typically involve contact with the flow. Furthermore, measurements at several sections are needed for the kinematic flow analysis. Ground penetrating radar may offer a non-contact means for measuring cross sections [*Spicer et al.*, 1997], although measurements at multiple sections could still be difficult at some sites. Other approaches for dis-

charge estimation, which only require a single cross section, might also be explored. For example, *Costa et al.* [2000] estimated depth-averaged velocity directly from surface velocity using an assumed vertical profile relationship. We are actively involved in experimental research to investigate profile relationships for noncontact discharge estimation. Still, whenever information on two- or three-dimensional flow patterns is desired, multiple cross sections are needed to fully utilize the flow field information obtained by video imagery.

[35] Although the video imagery approach for discharge estimation was demonstrated for low flows, the approach may be best suited to other applications. For example, the approach has significant potential for flood flow estimation at ungaged sites. Video equipment could be used to record flooding as the opportunity arises, and post-event surveys of the site would then be made to obtain cross section information for the kinematic flow analysis. Since measurements can be made quickly and with little site preparation, a crew can respond quickly in urgent situations, such as a flash flood. The video imagery techniques could also be very effective in situations where it is unsafe or impossible to obtain field measurements with conventional methods, such as near bridges, river training structures (e.g., chevron and wing dikes), power houses, and spillways. Another potential application is the measurement of flows from a remote site. Since the video-based system uses inexpensive equipment to make digital images of the flow surface, a camera and a system for transmitting the images could be placed at a site, permitting frequent measurements under varying flow conditions. Still, in such applications, it is important to recognize the potential for errors that exists whenever a cross section survey is not made at the time of the velocity measurement. As an example, a post-event survey may not accurately represent the channel bathymetry at the time of the video measurement due to the scour and deposition that can occur over the course of flood event.

[36] Another exciting aspect of this approach is that the kinematic flow analysis provides information on two- and three-dimensional flow components, in addition to estimates of total river discharge. Even without cross-sectional information for discharge estimation, the video-based approach could be very helpful in visualizing complex flow patterns, such as secondary currents, in situations where the flow is inaccessible by other means. This new information on two- and three-dimensional flows could be valuable to scientific and engineering investigations of channel stabilization, bank erosion, stream and wetland ecology, stream corridor restoration, or flood impacts on river geomorphology and riverine habitat. Still, additional experiments are needed to gather appropriate data to evaluate estimates of these high-order flow characteristics.

[37] **Acknowledgments.** The work was carried out with financial support from IIHR-Hydroscience and Engineering. Special thanks also go to V. C. Patel for his assistance and support in our research on environmental flow measurement using PIV. The authors would also like to thank Bill Eichinger, Mike Kundert, Li Chen, Jennifer Holman, Brian Nelson, and Paul Kucera, for their help in making measurements at Clear Creek, and Andrew McCoy, for his help in evaluating the current meter measurement errors. Finally, the authors would like to thank Randal Dinehart and Andrew Miller for their thorough reviews of an earlier version of this work. Their insightful comments and suggestions significantly improved this paper.

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